

App Development Project Report

Placement Portal Application

1. Student Details

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About Me:

I am a BE Mechatronics student with a strong interest in software development, robotics, and building real-world applications. I enjoy solving practical problems using technology and developing scalable systems.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a web-based application that enables efficient management of campus placements by handling **company registrations**, **student applications**, and **placement drives** in a centralized system.

Approach:

The application was developed using Flask as the backend framework with SQLite as the database. It supports role-based access for Admin, Company, and Student users. The system ensures proper approval workflows, job posting, and application tracking.

3. AI/LLM Declaration

I used **ChatGPT** for **guidance**, **debugging**, and **understanding implementation logic**.

The extent of AI/LLM usage is approximately **20–25%**, limited to **assistance** and **reference**.

All core implementation, logic design, and integration were done independently.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend web framework
SQLite	Local database storage
Jinja2	Template engine for dynamic HTML
HTML	Frontend structure
Bootstrap	UI styling and responsiveness

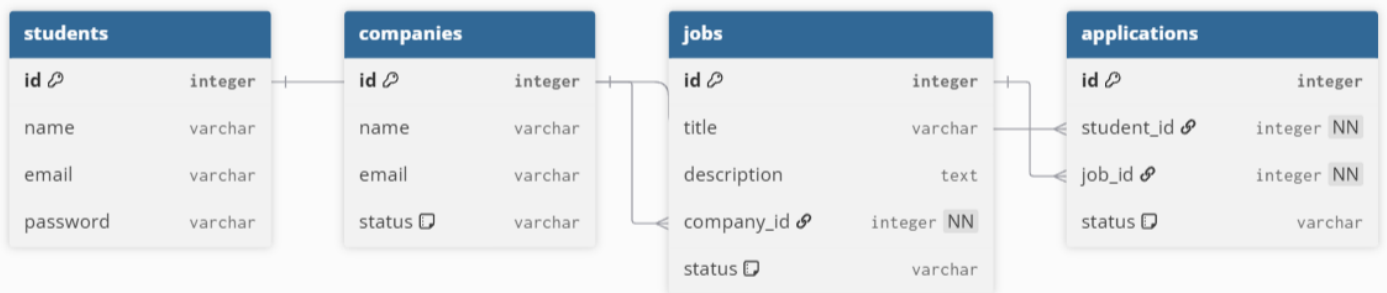
5. Database Schema / ER Diagram

Tables:

1. **Students** — stores student details (id, name, email, password)
2. **Companies** — stores company details and approval status
3. **Jobs** — contains job postings created by companies
4. **Applications** — tracks student applications

Relationships:

- One-to-Many → **Company** → **Jobs**
- One-to-Many → **Student** → **Applications**
- One-to-Many → **Job** → **Applications**



6. API Resource Endpoints

Endpoint	Method	Description
/login	POST	Authenticate user
/register	GET/POST	Register student/company
/admin	GET	Admin dashboard
/approve/	GET	Approve company
/company/	GET	Company dashboard
/create_drive/	POST	Create job
/apply/	POST	Apply for job

YAML API Definition File:

Included separately in the submission ZIP as [api.yaml](#).

7. Architecture and Features

Architecture Overview:

- **app.py** – main application logic and routing
- **templates/** – HTML pages for UI
- **static/** – styling files
- **database.db** – SQLite database created programmatically

Implemented Features:

- Role-based authentication system
- Admin approval for companies
- Job posting and management by companies
- Job application system for students
- Application status tracking
- Prevention of duplicate applications
- Display of only approved jobs

Additional Features:

- Multiple dashboards for each role
- Clean UI using Bootstrap
- Structured database design

8. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1rjEtmZa6PXxUQot9baheObbvtu5K2rBu/view?usp=drivesdk>